

YUHENG (AARON) DING

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EDUCATION

Carnegie Mellon University

Master of Science in Intelligent Information Systems (ML/NLP @ SCS LTI)
QPA: 4.21/4.0

Pittsburgh, PA

August 2025 - December 2026

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science, Bachelor of Science in Applied Mathematics
GPA: 3.88/4.0, Cum Laude

Los Angeles, CA

September 2021 - June 2025

EXPERIENCE

Google

Software Engineering Intern

Los Angeles, CA

May 2026 - August 2026

- Engineered a logging pipeline across YouTube's **C++** recommendation engine, frontend servers, and **Go** data warehouse to capture candidate ranking and sources for **80M+** daily suggestions to improve recommendation quality.
- Developed agent skill to reliably generate and monitor E2E testing configs within an LLM-as-tester framework for YouTube messaging, driving a **30%** decrease in manual QA and a **68.4%** reduction in token consumption.
- Deployed **SQL** workflows to monitor recommendation filters, detecting **83%** zero-yield filters to guide future upgrades.
- Built a metric dashboard for **20+** core recommendation quality metrics to establish system baseline for ML integration.

UCLA-NLP | Mentor: Kai-Wei Chang, Nanyun Peng

Undergrad Researcher

Los Angeles, CA

June 2024 - June 2025

- [**CVPR 2025, Accepted** [Link](#)] Developed a comprehensive benchmark for evaluating visual language models (VLMs) on perception and reasoning quality in QA tasks; tested on **26 VLMs** to assess self-critique and correction abilities.
- Managed a team of 3 annotators and built a **Python**-based annotation tool to curate over **1,600** annotated samples.
- Implemented an agent-based critique generation method at each reasoning step, improving accuracy by **13.5%**.

VisionX LLC

Full-Stack Software Engineer Intern

San Jose, CA (Remote)

July 2023 - October 2023

- Collaborated with 4 engineers on developing an environmental sustainability e-commerce platform using **Django**.
- Implemented payment workflow with Stripe API and email order confirmations, reducing checkout latency by **30%**.
- Integrated **WebRTC/WebSocket** voice and video chat into the reservation system, cutting confirmation time by **60%**.

PROJECTS

DeepSeek Sparse Attention Kernel Optimization for B200 - AI Kernel Developer

January 2026 - April 2026

- Engineered a DeepSeek Sparse Attention (DSA) **CUDA** kernel with **CUTLASS** for **NVIDIA B200** utilizing a TMA-driven, double-buffered, warp-specialized pipeline, achieving **101.39x** speedup over PyTorch and **18x** gain over FlashInfer.
- Developed a **FP8** Top-K indexer kernel by fusing ReLU and head-weighted reductions directly into a single-pass **GEMM**, delivering **244.71x** speed up over PyTorch and outperforming FlashInfer by **2.1x**.

Perspective Aware Visual Question Answering - Researcher

September 2025 - December 2025

- Finetuned **Qwen2.5-VL** for spatial perspective-taking tasks using **SFT** and **RL-GRPO**, achieving **10%** higher accuracy.
- Developed a **ReAct** agentic spatial reasoning pipeline with **Python**-based calculation tools, improving zero-shot VLM accuracy on 3D orientation tasks by up to **21%**.

Pittsburgh & CMU QA System - Team Lead

October 2025 - November 2025

- Led a team of 3 in building an end-to-end **RAG** Question Answering system with multiple chunking granularities.
- Designed a Small-to-Big hybrid retriever weighted score fusion, combining sentence-level **BM25** score with paragraph-level semantic search via **MiniLM-L6-v2** embeddings, achieving **75%** accuracy under LLM evaluation.

High Performance URL Shortening Web Server - Tech Lead & Contributor

April 2025 - June 2025

- Led a team of 4 in building an async TCP server in **C++** with **Boost.Asio** with test-driven development; deployed on **GCP** using **Docker** and **Compute Engine**; automated CI/CD with **Cloud Build** for zero-downtime deployment.
- Designed application architecture for URL shortening service with **Redis** caching and **PostgreSQL** persistence, leveraging connection pooling to support **1k+** concurrent users.

SKILLS

Programming Languages: Java, Python, C++, C, SQL, CUDA, JavaScript, Bash, Go

Frameworks & Libraries: React, HTML, CSS, Django, CUTLASS, Spring Boot, Tomcat, PyTorch, NumPy

Databases & Data Stores: PostgreSQL, MySQL, SQLite, Redis, MongoDB

Cloud & DevOps: AWS, GCP (Google Cloud), Azure, Docker, Kubernetes, Git, Linux

Languages: English (Fluent), Mandarin (Fluent), Spanish (Basic)